

Ashwanth Senthilkumar

Software Engineer | Full-Stack & AI/ML Developer

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[LinkedIn](#) | [GitHub](#) | [LeetCode](#)

PROFESSIONAL SUMMARY

Final-year Computer Science Engineering student (AI/ML Honors, SREC) with demonstrated expertise in full-stack development and applied machine learning. Published first author at ICDAI 2025. Built end-to-end AI applications integrating FastAPI, ReactJS, and LLM APIs (OpenAI, Gemini). Solved 300+ LeetCode problems; IEEE CIS Student Chapter Chairperson. Seeking a software engineering internship or entry-level SDE/ML role where proven ability to deliver production-quality, AI-driven solutions.

INTERNSHIP EXPERIENCE

Qono Technologies | Machine Learning Intern

22 May 2025 – 30 June 2025

Coimbatore | India

- Engineered end-to-end machine learning pipelines, encompassing data preprocessing, feature engineering, and hyperparameter tuning to build scalable recommendation systems for movies and books using collaborative and content-based filtering, matrix decomposition, and TF-IDF vectorization.
- Developed a production ready IoT Malware Classification model by applying supervised learning algorithms (Random Forest, SVM) to network traffic and behaviour features, enabling automated threat detection across connected environments.

Issential Solutions | AI/ML Intern

1 December 2025 – 31 December 2025

Coimbatore | India

- Integrated Explainable AI (XAI) frameworks (LIME, SHAP) into classification pipelines to extract local and global feature attributions, while analyzing trade-offs in computational cost and explanation fidelity across multiple model architectures.
- Enhanced model transparency for non-technical stakeholders by translating complex black-box predictions into accessible visual reports (waterfall and force plots), successfully reducing trust barriers in decision-critical workflows.

EDUCATION

Sri Ramakrishnan Engineering College (SREC)

Bachelor of Engineering - Computer Science | CGPA: 7.94/10

Honors specialization in Artificial Intelligence & Machine Learning

Coimbatore, Tamil Nadu

August 2023 – Present

June 2025 - Present

Bharani Vidhyalaya Sr. Sec. School (CBSE)

HSC | Percentage: 80%

SLSC | Percentage: 75%

Karur, Tamil Nadu

June 2022 – May 2023

June 2020 – May 2021

SKILLS

Languages: Python, Java, JavaScript, R

Frontend: ReactJS, HTML5, CSS3

Backend & APIs: FastAPI, REST API Design, API Integration, JSON

AI/ML: TensorFlow, Keras, Scikit-learn, OpenCV, Pandas, NumPy, LLM Integration (OpenAI, Gemini), Hugging Face, Prompt Engineering, Feature Engineering, Model Evaluation

Databases: SQL, MySQL, PostgreSQL, MongoDB

DevOps & Tools: Git, GitHub, Docker, Jupyter Notebook, Tableau, Google Colab

Core Concepts: Data Structures & Algorithms, OOPs, DBMS, Cloud Computing (AWS, GCP Vertex AI), System Design, Agile/Scrum Development, MLOps, MLFlow

PROJECTS

GreenRoute-Carbon Optimized Travel Planner | Python • Scikit-learn • FastAPI • Streamlit | [GitHub](#)

- Engineered an ML regression model to predict CO₂ emissions per travel route using distance, traffic density, vehicle type, and fuel-efficiency features, achieving 84% prediction accuracy on held-out test data.
- Implemented a feature engineering pipeline processing geospatial and vehicle-type variables, improving model predictive performance by 95%.
- Built REST APIs using FastAPI and integrated a Streamlit dashboard for real-time route comparison.

Recipe Archaeologist - GenAI | Python • OpenCV • FastAPI • Gemini API • ReactJS | [GitHub](#)

- Architected a multimodal AI pipeline to reverse-engineer culinary recipes from food residue images, integrating computer vision (OpenCV) with LLM-based ingredient inference.
- Designed a rule-based molecular inference layer to derive ingredient probabilities from visual pattern analysis, enhancing recipe reconstruction interpretability.
- Developed full-stack solution with FastAPI backend and ReactJS UI, delivering end-to-end image-to-recipe output in under 10 seconds.
- Applied prompt engineering techniques to optimize LLM reasoning accuracy for ingredient identification in ambiguous visual contexts.

CERTIFICATION

• Machine Learning Specialization – Coursera Stanford University (Andrew Ng)	June 2025
• Deep Learning – NPTEL (IIT Ropar)	April 2025
• Introduction to Large Language Model – NPTEL (IIT Madras)	April 2026
• Social Network Analysis – NPTEL (IIT Hyderabad)	November 2025
• Citizen Data Science using Python Certification — Infosys Springboard	August 2025
• Career Essentials in GitHub Certification — GitHub & LinkedIn Learning	January 2025

PUBLICATION & ACTIVITIES

- **IEEE Computational Intelligence Society (SREC) | Student Chapter Chairperson** **2026 – Present**
Led chapter operations for 10 student members; organized 2 technical workshops, seminars, and professional development events throughout the academic year. Coordinated chapter inauguration ceremony, speaker sessions, and technical talks, increasing student participation in IEEE CIS programs.
- **Supply Chain Forecasting for Inventory Management | First Author • International Conference on Data Analytics and Insights (ICDAI 2025) • Paper Published**
Developed a forecasting model to optimize inventory replenishment across supply chain nodes using time-series analysis and ML regression techniques. Presented research findings at ICDAI 2025; paper published in Springer.